ZC416 • MATHEMATICAL FOUNDATIONS FOR ML / DS

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Unveiling the Matrix

Determinants, Eigenvalues, Eigenvectors and the Spectral Theorem

This companion is the in-depth, visual partner to Lecture 4. We begin with the two single-number "fingerprints" of a matrix — the **determinant** (how much a transformation scales space, and whether it is reversible) and the **trace** — then use the **characteristic equation** to hunt down **eigenvalues** and the **eigenvectors** they belong to. The journey peaks at the **Spectral Theorem** and the decomposition $A = Q\Lambda Q^{\top}$, before reaching out to **Hermitian** matrices and the **Cholesky** factorisation — the tools that quietly power PCA, SVD, simulation, and quantum mechanics.

A connected story — every idea here feeds the next.

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1 Determinants: the measure of a transformation's power

Before we hunt for the special directions of a matrix, we need a way to quantify what a transformation *does* to space as a whole. How much does it stretch or shrink? Does it flatten space into a lower dimension? That single, powerful number is the **determinant**.

What does a determinant tell us?

Imagine a matrix A acting on a unit square (in 2D) or a unit cube (in 3D).

- The **absolute value** $|\det(A)|$ is the factor by which **area** (2D) or **volume** (3D) is scaled. $\det(A) = 2$ doubles the area; $\det(A) = 0.5$ halves it.
- $\det(A) = 0$ is the **alarm bell**: the transformation squashes space into a lower dimension (a square flattened to a line, a cube to a plane). The matrix is then **singular** — non-invertible — because information has been destroyed.
- The **sign** of $\det(A)$ records orientation: positive preserves it; negative **flips** space, like looking in a mirror.

In plain words. Think of rolling out pizza dough. If you press and stretch it so it covers twice the area, that "$\times 2$" is the determinant. If you flatten the dough completely into a thin crease, its area becomes zero — and you can never roll it back into the original shape. That flattening is exactly what $\det(A) = 0$ means. In one number, the determinant captures the "oomph" of a matrix.

1.1 Defining determinants: minors and cofactors

The geometric picture is the intuition; the formula below is how we actually compute. For an $n \times n$ matrix A the determinant is built *recursively* from smaller determinants.

Definition 1.1: Minor and cofactor

Let A be an $n \times n$ matrix.

- The **minor** M_{ij} of the entry a_{ij} is the determinant of the $(n-1) \times (n-1)$ submatrix left after deleting row i and column j.
- The **cofactor** is $C_{ij} = (-1)^{i+j} M_{ij}$. The factor $(-1)^{i+j}$ creates a checkerboard pattern of signs.

Definition 1.2: Determinant by cofactor expansion

For an $n \times n$ matrix A, expanding along *any* row i or *any* column j gives the same value:

$$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} = \sum_{i=1}^{n} a_{ij} C_{ij}.$$

Base cases. $\det[a] = a$, and for a 2×2 matrix $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

Example 1.1: Expanding a 3×3 determinant

Let $A = \begin{pmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{pmatrix}$ and expand along the first row, $\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$:

$$C_{11} = (+1)\det\begin{pmatrix} -4 & 3 \\ 4 & -2 \end{pmatrix} = (8 - 12) = -4,$$

$$C_{12} = (-1)\det\begin{pmatrix} -2 & 3 \\ 5 & -2 \end{pmatrix} = -(4 - 15) = 11,$$

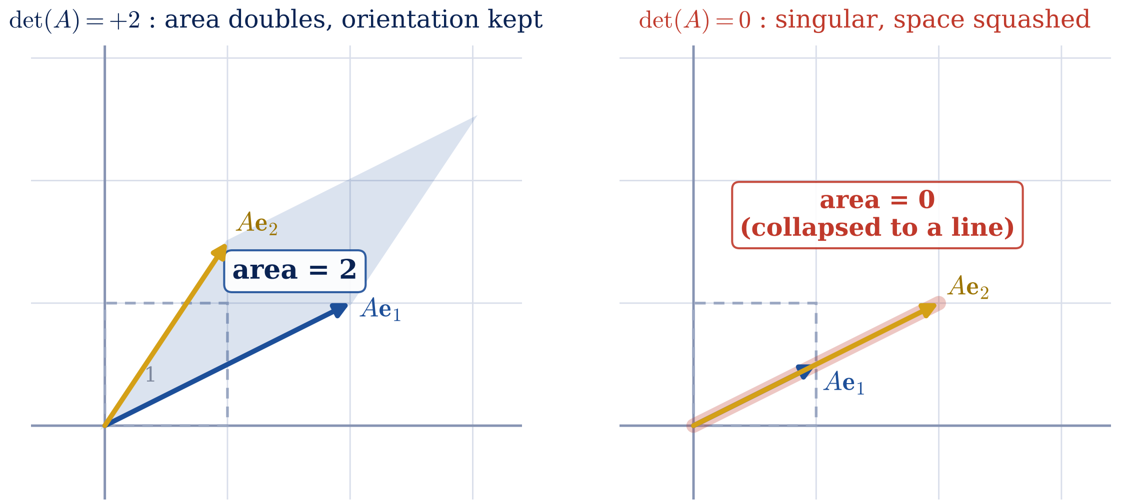
$$C_{13} = (+1)\det\begin{pmatrix} -2 & -4 \\ 5 & 4 \end{pmatrix} = (-8 + 20) = 12.$$

Therefore $\det(A) = 3(-4) + 1(11) + 0(12) = -12 + 11 + 0 = \boxed{-1}$.

Choose the lazy row or column

Expand along the row or column with the **most zeros**. The $a_{13} = 0$ term above cost us nothing — every zero entry removes an entire cofactor from the work.

1.2 The geometric meaning of the determinant



The determinant as an area-scaling factor. ***Left:*** *the columns of $A = \begin{pmatrix} 2 & 1 \\ 1 & 1.5 \end{pmatrix}$ are the images $A\mathbf{e}_1, A\mathbf{e}_2$ of the standard basis; the unit square (dashed) becomes a parallelogram of area $|\det(A)| = 2$.* ***Right:*** *when the columns become linearly dependent the parallelogram collapses to a line, $\det = 0$, and the matrix is singular.*

The columns of A are precisely the images of the standard basis vectors, $A\mathbf{e}_j$. So the parallelogram (2D) or parallelepiped (3D) they span has signed volume equal to $\det(A)$. This single picture explains a remarkable amount:

- **Linear dependence $\Leftrightarrow$ zero determinant.** If the columns are linearly dependent they cannot span the full space; the parallelepiped is "flat" with zero volume, i.e. $\det(A) = 0$.
- **Invertibility.** A is invertible *if and only if* $\det(A) \neq 0$. If the transformation collapses space, there is no way to uniquely undo it.

The properties we will lean on

$\det(AB) = \det(A)\det(B)$	scaling factors of composed maps multiply
$\det(A^\top) = \det(A)$	transposing does not change the volume scaling
$\det(A^{-1}) = 1/\det(A)$	the inverse scales volume by the reciprocal
A triangular $\Rightarrow \det(A) = \prod_i a_{ii}$	a huge computational shortcut

Theorem 1.1: Full rank $\Leftrightarrow$ non-zero determinant

An $n \times n$ matrix A has rank n (full rank) if and only if $\det(A) \neq 0$.

Why. Gaussian elimination reduces A to an upper-triangular U using s row interchanges, and $\det(A) = (-1)^s \det(U) = (-1)^s \prod_i U_{ii}$. The matrix has full rank exactly when all pivots U_{ii} are non-zero, which is exactly when this product — and hence $\det(A)$ — is non-zero. **Hold on to this:** an eigenvalue is born precisely where a *shifted* matrix $A - \lambda I$ loses full rank.

2 The trace: a matrix's hidden sum

The determinant's understated sibling is the **trace**. It looks almost too simple to be useful — you just add up the diagonal — yet it secretly equals the *sum of the eigenvalues*, a fact we will cash in shortly. Think of it as a doctor taking a quick pulse: one easy number that already tells you a lot about the patient.

Definition 2.1: Trace

The trace of an $n \times n$ matrix A is the sum of its diagonal entries,

$$\mathrm{tr}(A) = \sum_{i=1}^{n} a_{ii}.$$

Properties (all short to prove)

$$\mathrm{tr}(A+B) = \mathrm{tr}(A) + \mathrm{tr}(B) \qquad \mathrm{tr}(\alpha A) = \alpha\, \mathrm{tr}(A)$$
$$\mathrm{tr}(I_n) = n \qquad \mathrm{tr}(AB) = \mathrm{tr}(BA)$$

The last, "cyclic" identity $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ is the workhorse behind countless machine-learning derivations (for instance gradients of $\mathrm{tr}(X^\top AX)$).

A teaser we prove in Section 5

When the characteristic polynomial is expanded, the trace and determinant fall out as its first and last coefficients:

$$p_A(\lambda) = \det(A - \lambda I) = (-1)^n \lambda^n + (-1)^{n-1}\,\mathrm{tr}(A)\,\lambda^{n-1} + \cdots + \det(A).$$

Equivalently, $\mathrm{tr}(A) = \sum_i \lambda_i$ and $\det(A) = \prod_i \lambda_i$: two of the simplest possible summaries of the eigenvalues, readable straight off the matrix.

3 The essence of a matrix: eigenvalues and eigenvectors

Every square matrix A describes a transformation: it takes a vector $\mathbf{x}$ to a new vector $A\mathbf{x}$, stretching, rotating, shearing or reflecting space. Within that motion, are there any vectors whose *direction* survives untouched — vectors that are merely scaled? Yes: these are the **eigenvectors**, the "grain" or natural axes of the matrix, and the scale factors are the **eigenvalues**.

Finding the grain of a transformation

- **Spinning globe.** The rotation axis (through the poles) does not move — an eigenvector with eigenvalue 1. Every other point swings around it.
- **Stretching dough.** Pulling along one direction and squeezing another: the pull direction is an eigenvector with $\lambda > 1$, the squeeze direction one with $0 < \lambda < 1$.
- **Shearing a deck of cards.** Push the top sideways: horizontal vectors stay horizontal ($\lambda = 1$), while everything else tilts.

Definition 3.1: Eigenvector and eigenvalue

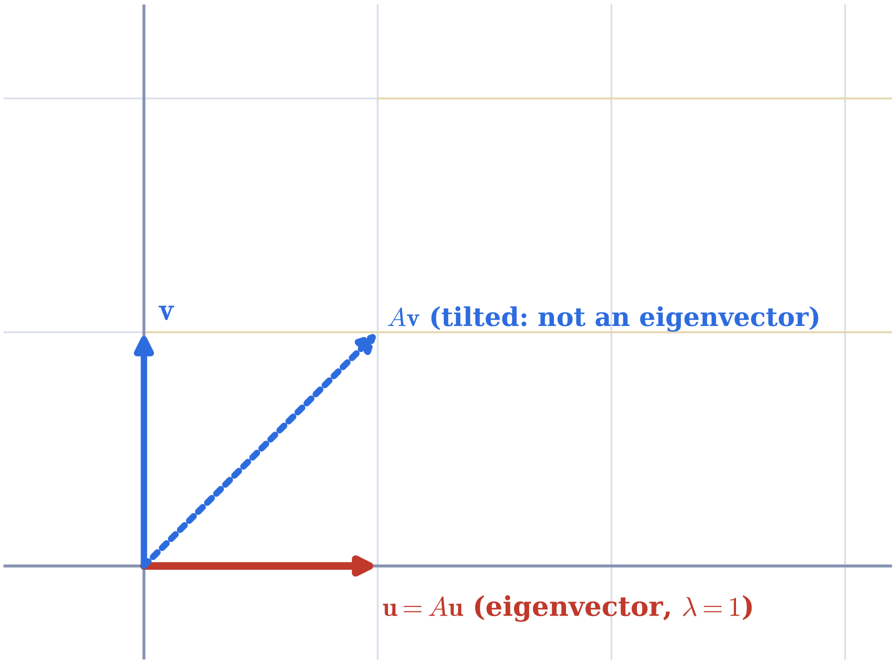
A **nonzero** vector $\mathbf{x}$ is an eigenvector of A if, for some scalar λ,

$$A\mathbf{x} = \lambda\mathbf{x}.$$

The scalar λ is the corresponding eigenvalue. A positive λ scales in the same direction; a negative λ flips the direction (a reflection plus scaling).

Note. The zero vector satisfies $A\mathbf{0} = \lambda\mathbf{0}$ for every λ, but it points nowhere, so it is explicitly excluded from being an eigenvector.

A shear has just one eigen-direction



A shear $A = \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$. The red vector $\mathbf{u}$ along the x-axis is fixed ($A\mathbf{u} = \mathbf{u}$, so $\lambda = 1$): it is an eigenvector. The blue vector $\mathbf{v}$ is tilted by the shear, so it is not *an eigenvector. A shear has only this one eigen-direction.*

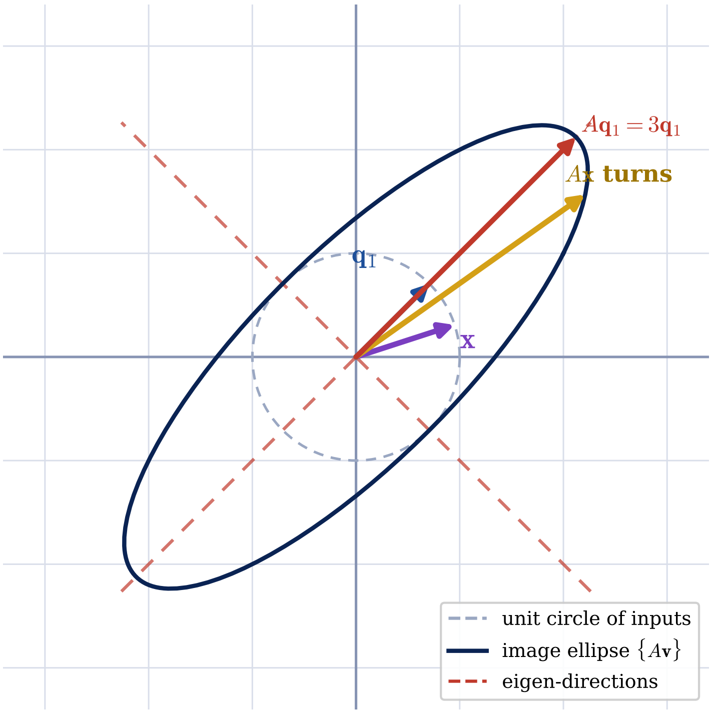
Example 3.1: Verifying eigenvectors by hand

Let $A = \begin{pmatrix}1 & 6\\ 5 & 2\end{pmatrix}$, $\mathbf{u} = \begin{pmatrix}6\\ -5\end{pmatrix}$, $\mathbf{v} = \begin{pmatrix}3\\ -2\end{pmatrix}$.

$$A\mathbf{u} = \begin{pmatrix}1(6) + 6(-5)\\ 5(6) + 2(-5)\end{pmatrix} = \begin{pmatrix}-24\\ 20\end{pmatrix} = -4\begin{pmatrix}6\\ -5\end{pmatrix} = -4\,\mathbf{u}.$$

So $\mathbf{u}$ *is* an eigenvector with $\lambda = -4$. But $A\mathbf{v} = \begin{pmatrix}-9\\ 11\end{pmatrix}$, which is not a scalar multiple of $\mathbf{v}$ (it would need $\lambda = -3$ *and* $\lambda = -\frac{11}{2}$ at once). So $\mathbf{v}$ is **not** an eigenvector.

Eigenvectors keep direction; everything else turns



The action of the symmetric matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. The unit circle of inputs (dashed) is mapped to an ellipse (navy). Along the eigen-directions (red dashed) a vector is only scaled — here $A\mathbf{q}_1 = 3\mathbf{q}_1$ — while a generic vector $\mathbf{x}$ has its direction changed.

4 How to find eigenvalues and eigenvectors

Dragging vectors around builds intuition, but we need a systematic *method*. The trick is to turn the geometric question into an algebra problem: finding the roots of a polynomial.

A helpful picture. An eigenvalue is like the exact musical note that makes a wine glass ring. Most frequencies do nothing; but at one special pitch the glass resonates and can even shatter. In the same way, most values of λ leave $A - \lambda I$ perfectly solid and invertible — but at the special eigenvalues it "goes limp" and collapses space. Our job is to find exactly those values.

4.1 The characteristic equation

Starting from the definition and collecting terms on one side,

$$A\mathbf{x} = \lambda\mathbf{x} \implies A\mathbf{x} - \lambda I\mathbf{x} = \mathbf{0} \implies (A - \lambda I)\mathbf{x} = \mathbf{0}.$$

We need a **nonzero** solution $\mathbf{x}$. A square matrix maps a nonzero vector to $\mathbf{0}$ only when it *collapses space* — that is, only when it is singular — and we learned in Section 1 that this happens exactly when its determinant vanishes.

Why the determinant must be zero

Write $M = A - \lambda I$. If M is invertible ($\det M \neq 0$) the only solution of $M\mathbf{x} = \mathbf{0}$ is the trivial $\mathbf{x} = \mathbf{0}$, which we reject. For a genuine eigenvector to exist, M must collapse space — mapping a whole line or plane to $\mathbf{0}$ — and that requires $\det(A - \lambda I) = 0$.

Definition 4.1: The characteristic equation

The eigenvalues of A are the solutions λ of

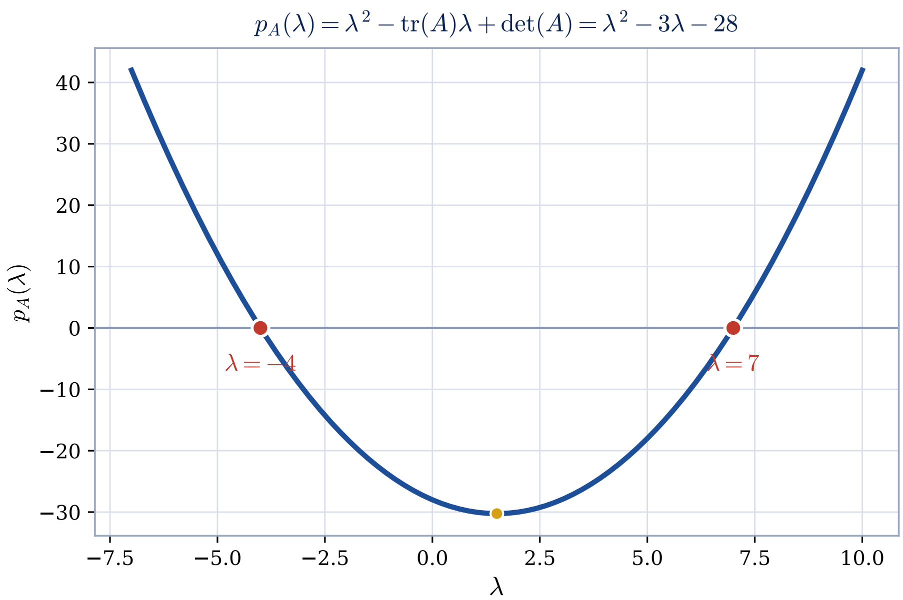
$$\det(A - \lambda I) = 0.$$

The left-hand side is a degree-n polynomial in λ, the **characteristic polynomial** $p_A(\lambda)$; its roots are the eigenvalues. For each eigenvalue, the nonzero solutions of $(A - \lambda I)\mathbf{x} = \mathbf{0}$ form the **eigenspace** E_λ.

What the coefficients are. Expanding $\det(A - \lambda I)$ for a general $n \times n$ matrix, the only term producing the highest powers of λ is the diagonal product $\prod_i (a_{ii} - \lambda)$. Reading off its top two coefficients (and noting every other cofactor contributes a lower power) gives

$$p_A(\lambda) = (-1)^n \lambda^n + (-1)^{n-1}\,\mathrm{tr}(A)\,\lambda^{n-1} + \cdots + \det(A),$$

where the constant term is $p_A(0) = \det(A)$. So **the trace and determinant are literally the top and bottom coefficients of the characteristic polynomial.**



The characteristic polynomial of $A = \left(\begin{smallmatrix}1 & 6\\ 5 & 2\end{smallmatrix}\right)$ *is* $p_A(\lambda) = \lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = \lambda^2 - 3\lambda - 28$. *Its roots, where the parabola crosses zero, are the eigenvalues* $\lambda = 7$ *and* $\lambda = -4$.

Example 4.1: Finding eigenvalues and eigenspaces

For $A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$:

$$\det(A - \lambda I) = \det\begin{pmatrix} 1-\lambda & 6 \\ 5 & 2-\lambda \end{pmatrix} = (1-\lambda)(2-\lambda) - 30 = \lambda^2 - 3\lambda - 28 = (\lambda-7)(\lambda+4).$$

So $\lambda_1 = 7,\ \lambda_2 = -4$. For each, solve $(A - \lambda I)\mathbf{x} = \mathbf{0}$:

$$\lambda_1 = 7: \begin{pmatrix} -6 & 6 \\ 5 & -5 \end{pmatrix}\mathbf{x} = \mathbf{0} \Rightarrow x_1 = x_2, \quad E_7 = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\};$$

$$\lambda_2 = -4: \begin{pmatrix} 5 & 6 \\ 5 & 6 \end{pmatrix}\mathbf{x} = \mathbf{0} \Rightarrow 5x_1 + 6x_2 = 0, \quad E_{-4} = \operatorname{span}\left\{\begin{pmatrix} 6 \\ -5 \end{pmatrix}\right\}.$$

4.2 A shortcut for triangular matrices

Theorem 4.1: Eigenvalues of a triangular matrix

The eigenvalues of a triangular matrix (upper or lower) are exactly its diagonal entries.

Why. If A is triangular then so is $A - \lambda I$, and the determinant of a triangular matrix is the product of its diagonal. Thus $\det(A - \lambda I) = (a_{11} - \lambda)(a_{22} - \lambda)\cdots(a_{nn} - \lambda) = 0$, whose solutions are simply $\lambda = a_{11}, a_{22}, \ldots, a_{nn}$. No computation required — a clean consequence of the determinant properties.

5 Trace, determinant and the eigenvalues

We promised that the trace and determinant summarise the eigenvalues. Here is the precise statement.

Theorem 5.1: Sum and product of eigenvalues

For any $n \times n$ matrix A with eigenvalues $\lambda_1, \ldots, \lambda_n$ (counted with multiplicity),

$$\sum_{i=1}^{n} \lambda_i = \operatorname{tr}(A), \qquad \prod_{i=1}^{n} \lambda_i = \det(A).$$

Why. Because the λ_i are the roots of the characteristic polynomial, we may factor it as $p_A(\lambda) = \prod_{i=1}^{n}(\lambda_i - \lambda)$. Expanding, the coefficient of λ^{n-1} is $(-1)^{n-1}\sum_i \lambda_i$; comparing with the direct expansion from Section 4, where that coefficient is $(-1)^{n-1}\operatorname{tr}(A)$, gives $\sum_i \lambda_i = \operatorname{tr}(A)$. Setting $\lambda = 0$ in the factored form gives $\prod_i \lambda_i = p_A(0) = \det(A)$.

A free sanity check

For $A = \left(\begin{smallmatrix} 1 & 6 \\ 5 & 2 \end{smallmatrix}\right)$ we found $\lambda = 7, -4$. Indeed $7 + (-4) = 3 = \operatorname{tr}(A)$ and $7 \cdot (-4) = -28 = \det(A)$. Whenever you compute eigenvalues, this pair of checks catches arithmetic slips instantly.

6 The Spectral Theorem: the crown jewel for symmetric matrices

We now reach one of the most useful results in all of linear algebra, and one that is especially dear to data science: nearly every matrix we build from data — a covariance matrix, or a Gram matrix $A^\top A$ — is symmetric.

Think of wood grain. A plank of wood splits cleanly along its grain and resists across it. A symmetric matrix behaves the same way: it has a natural "grain" — a set of perpendicular directions — along which it only stretches or shrinks, never twisting the wood. The Spectral Theorem is the precise statement that this clean grain always exists.

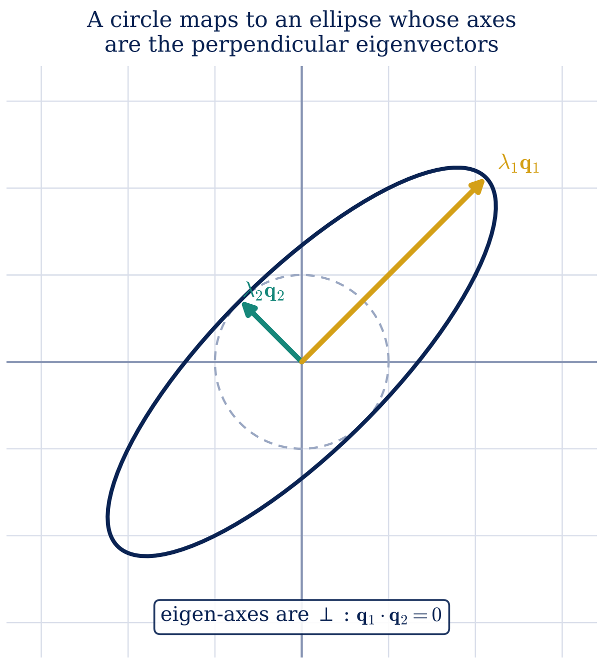
Definition 6.1: Symmetric matrix

A square matrix is **symmetric** if $A = A^\top$, i.e. $a_{ij} = a_{ji}$. It is mirrored across its main diagonal, e.g. $S = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 5 & 3 \\ 0 & 3 & -4 \end{pmatrix}$.

Theorem 6.1: Spectral Theorem for real symmetric matrices

If $A \in \mathbb{R}^{n\times n}$ is symmetric, then:

1. all eigenvalues are **real**;
2. eigenvectors of **distinct** eigenvalues are **orthogonal**;
3. there is an **orthonormal basis** of $\mathbb{R}^n$ consisting of eigenvectors of A — so A is always diagonalisable, and beautifully so, by an orthogonal matrix.



A symmetric matrix has perpendicular eigen-axes. The unit circle maps to an ellipse whose principal axes lie exactly along the orthogonal eigenvectors $\mathbf{q}_1, \mathbf{q}_2$*, each stretched by its eigenvalue.*

Why real. Let $A\mathbf{x} = \lambda\mathbf{x}$ and premultiply by the conjugate-transpose $\mathbf{x}^H$: then $\mathbf{x}^H A\mathbf{x} = \lambda\,\mathbf{x}^H\mathbf{x}$. The left side equals its own conjugate (because A is symmetric/Hermitian), so it is real; and $\mathbf{x}^H\mathbf{x} = \|\mathbf{x}\|^2 > 0$ is real and positive. Hence λ is real.

Why orthogonal. For $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$ with $\lambda \neq \mu$,

$$\lambda\,\mathbf{y}^H\mathbf{x} = \mathbf{y}^H A\mathbf{x} = (A\mathbf{y})^H\mathbf{x} = \mu\,\mathbf{y}^H\mathbf{x} \implies (\lambda-\mu)\mathbf{y}^H\mathbf{x} = 0,$$

and since $\lambda \neq \mu$ we must have $\mathbf{y}^H\mathbf{x} = 0$: the eigenvectors are perpendicular.

6.1 Eigendecomposition: $A = Q\Lambda Q^\top$

Because a symmetric matrix has a full orthonormal set of eigenvectors, we can **factor** it:

$$A = Q\,\Lambda\,Q^\top,$$

where Q has the orthonormal eigenvectors as columns (so $Q^{-1} = Q^\top$) and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

In plain words. Imagine a crooked photo on the wall that you want to enlarge. You first *turn* it straight (that is $Q^\top$), then *stretch* it neatly along its now-horizontal and vertical edges (that is Λ), then *turn it back* to its original tilt (that is Q). A messy-looking transformation is really just turn, stretch, turn back.

Example 6.1: Eigendecomposition of a symmetric matrix

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. From $\det(A - \lambda I) = (2-\lambda)^2 - 1 = (\lambda-1)(\lambda-3)$ we get $\lambda_1 = 3$, $\lambda_2 = 1$. Solving for eigenvectors and normalising:

$$\mathbf{q}_1 = \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{q}_2 = \tfrac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad Q = \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

A quick check confirms $\mathbf{q}_1 \cdot \mathbf{q}_2 = 0$ and

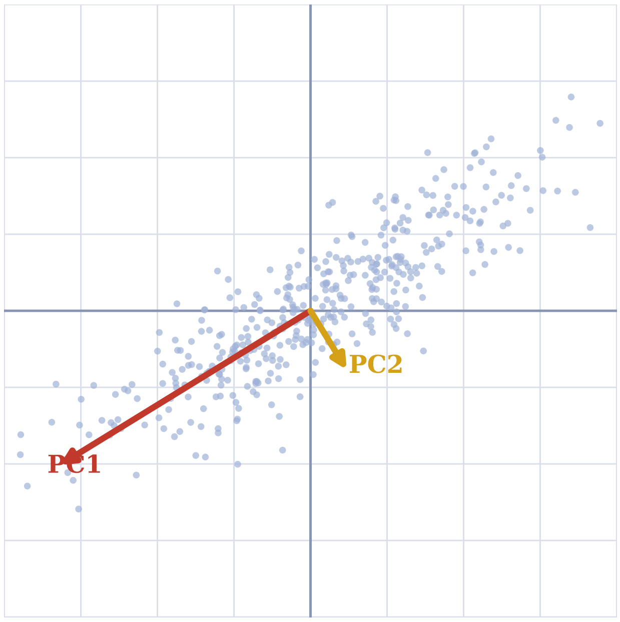
$$Q\Lambda Q^\top = \tfrac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = A. \checkmark$$

Reading $A = Q\Lambda Q^\top$ from right to left turns one "complicated" transformation into three simple moves applied to $\mathbf{x} = (1, 0.5)$. ***(1)*** *$Q^\top$ rotates into the eigenbasis.* ***(2)*** *Λ scales along those axes by the eigenvalues ($\times 3$ and $\times 1$).* ***(3)*** *Q rotates back, yielding $A\mathbf{x} = (2.5, 2.0)$.*

6.2 The big picture: why eigendecomposition matters

Eigenvalues, eigenvectors, and the decomposition of symmetric matrices are not curiosities — they are the engine room of modern data analysis.

PCA: principal components are eigenvectors of the covariance matrix



Principal Component Analysis. The principal components are the eigenvectors of the (symmetric) covariance matrix; their eigenvalues measure the variance captured along each. Keeping the largest few compresses data with minimal loss.

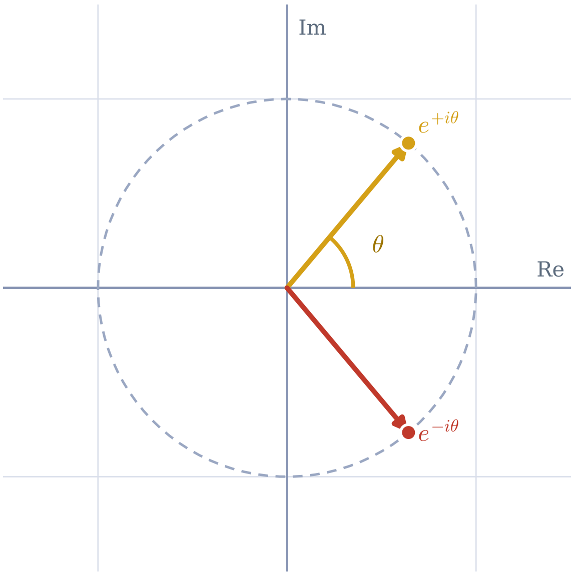
- **Principal Component Analysis (PCA).** The eigenvectors of the covariance matrix point along directions of maximum variance; their eigenvalues say how much variance each carries. Keeping the top eigenvectors gives optimal low-dimensional projections. The Spectral Theorem guarantees these directions are orthogonal and the eigenvalues real.
- **Singular Value Decomposition (SVD).** Built from the eigenvectors of the symmetric Gram matrices $A^\top A$ and $AA^\top$ (always positive semidefinite), SVD generalises eigendecomposition to *any* matrix — the backbone of recommender systems, latent semantic analysis, and low-rank compression.
- **Differential equations.** Solutions of $\frac{d\mathbf{y}}{dt} = A\mathbf{y}$ are combinations of $e^{\lambda t}\mathbf{v}$; the *signs of the eigenvalues* decide whether the system decays, oscillates, or blows up.
- **Quadratic forms and ellipsoids.** For symmetric A, $\mathbf{x}^\top A\mathbf{x} = c$ defines an ellipsoid whose axes are the eigenvectors and whose semi-axis lengths scale with $1/\sqrt{\lambda_i}$ — central to optimisation, inertia tensors, and confidence regions.

7 Complex vectors and Hermitian matrices

The eigenvector picture said a matrix only rescales an eigenvector along its own line. So what about a pure **rotation**, which turns *every* real direction? The resolution is to let the eigenvalues be **complex**.

Picture a merry-go-round. As it spins, not a single horse keeps pointing the same way — every direction is carried around. So there is no real eigenvector to find. Complex numbers come to the rescue because they are built to describe rotation: the eigenvalue $e^{i\theta}$ literally encodes "turn by the angle θ".

Rotation eigenvalues sit on the unit circle: $\lambda = e^{\pm i\theta}$



The eigenvalues of a rotation matrix R_θ live on the unit circle of the complex plane at $e^{\pm i\theta}$, always arriving as a conjugate pair. They encode "rotate by θ, scale by 1".

The rotation $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ has characteristic polynomial $\lambda^2 - 2\cos\theta\,\lambda + 1$, with roots $\lambda = \cos\theta \pm i\sin\theta = e^{\pm i\theta}$. Unless $\theta = 0$ or π, these are genuinely complex and occur in conjugate pairs. A complex eigenvalue $re^{\pm i\theta}$ encodes a *rotation by θ together with scaling by r*.

Definition 7.1: Conjugate transpose and length

For complex vectors the real recipe $\mathbf{x}^\top\mathbf{x}$ is no longer a sensible length (for $\mathbf{x} = (1+i,\, 2+i)^\top$ it gives $6 + 3i$, not even real). The repair is the **conjugate-transpose** $\mathbf{x}^H = \overline{\mathbf{x}}^\top$, giving

$$\mathbf{x}^H\mathbf{x} = \overline{x_1}x_1 + \cdots + \overline{x_n}x_n = |x_1|^2 + \cdots + |x_n|^2 = \|\mathbf{x}\|^2 \geq 0.$$

Hermitian: symmetric, upgraded

A matrix is **Hermitian** if $A^H = A$ — the complex generalisation of $A^\top = A$. For example $A = \left(\begin{smallmatrix} 1 & 3-i \\ 3+i & 4 \end{smallmatrix}\right)$ satisfies $A^H = A$. The Spectral Theorem extends perfectly: Hermitian matrices have **real eigenvalues** and **orthogonal eigenvectors**. This is exactly why quantum mechanics represents every physical observable (energy, momentum, spin) by a Hermitian operator: measured values must be real.

8 The Cholesky decomposition

Our last factorisation specialises to symmetric, **positive-definite** matrices (all eigenvalues positive). Such a matrix has a particularly tidy "square root". Just as $9 = 3 \times 3$ lets us write $\sqrt{9} = 3$, the Cholesky factor L is a kind of square root of a matrix: multiply it by its own mirror image and you get A back. And just like ordinary square roots, it only works for "positive" inputs — here, positive-definite matrices.

Theorem 8.1: Cholesky factorisation

A symmetric positive-definite matrix A can be written

$$A = L\,L^\top,$$

where L is lower-triangular with positive diagonal entries.

The entries of L are found one at a time, top-left to bottom-right:

$$\ell_{11} = \sqrt{a_{11}}, \quad \ell_{21} = \frac{a_{21}}{\ell_{11}}, \quad \ell_{22} = \sqrt{a_{22} - \ell_{21}^2}, \; \ldots$$

Each diagonal entry is a genuine square root, which stays real only when the matrix is positive-definite — so Cholesky doubles as a **positive-definiteness test**.

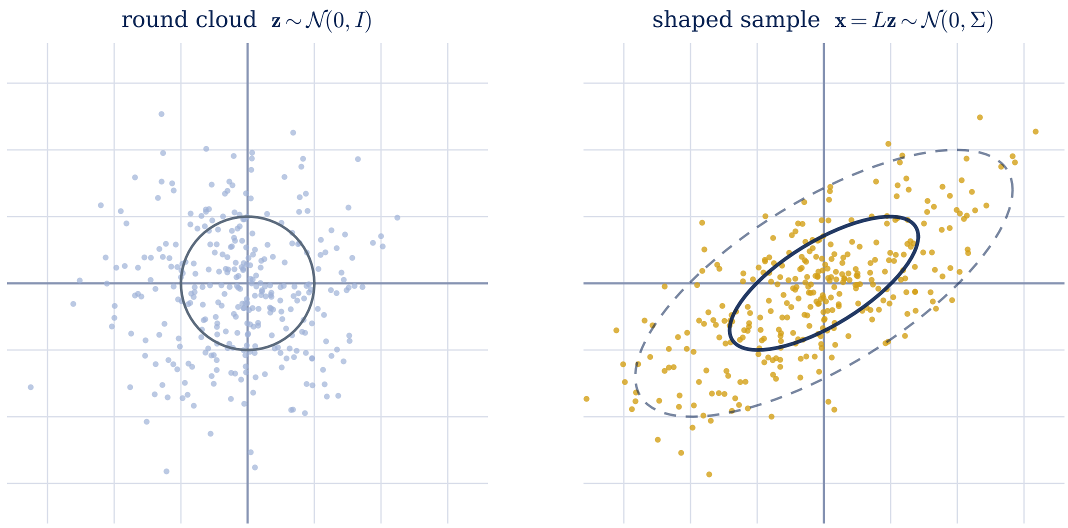
Example 8.1: A 2×2 Cholesky

For $A = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$: $\ell_{11} = \sqrt{4} = 2$, $\ell_{21} = 2/2 = 1$, $\ell_{22} = \sqrt{3-1} = \sqrt{2}$. Thus

$$L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}, \qquad LL^\top = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} = A.\checkmark$$

The killer application: sampling a Gaussian

A multivariate Gaussian is governed by a symmetric, positive-definite **covariance matrix** Σ. To draw a correlated sample: **(1)** factor $\Sigma = LL^\top$; **(2)** draw an easy *uncorrelated* standard-normal vector $\mathbf{z}$; **(3)** set $\mathbf{x} = L\mathbf{z}$ — now $\mathbf{x}$ has exactly covariance Σ. The factor L is the lens that bends a featureless round cloud into the correlated ellipse the data actually has.



*Sampling with Cholesky. **Left:** an uncorrelated standard-normal cloud $\mathbf{z} \sim \mathcal{N}(0, I)$. **Right:** after $\mathbf{x} = L\mathbf{z}$ the cloud is stretched and tilted into $\mathcal{N}(0, \Sigma)$; the navy curves are the 1- and 2-standard-deviation covariance ellipses.*

9 A look beyond: further properties and considerations

9.1 Diagonalisability

The factorisation $A = PDP^{-1}$ (with D diagonal) is called **diagonalisation**, and requires n linearly independent eigenvectors to fill the columns of P.

- The Spectral Theorem guarantees *every* real symmetric matrix is diagonalisable, and orthogonally so ($P = Q$, $P^{-1} = Q^\top$).
- A matrix with n **distinct** eigenvalues is always diagonalisable, since eigenvectors of distinct eigenvalues are independent.
- With **repeated** eigenvalues it might fail. The shear $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has the single eigenvalue $\lambda = 1$ (algebraic multiplicity 2) but only a one-dimensional eigenspace $\operatorname{span}\{(1,0)^\top\}$ — too few independent eigenvectors to diagonalise.

9.2 Complex eigenvalues from real matrices

A real but non-symmetric matrix can have complex eigenvalues, and they always appear in conjugate pairs $\lambda = a \pm bi$. Geometrically (as with the rotation above) a complex pair $re^{\pm i\theta}$ signals a rotation-and-scaling acting within an invariant plane — a motion with no fixed real direction.

The power of abstraction

Eigenvalues and eigenvectors reach far beyond matrices of numbers. They reappear as the eigenfunctions of differential operators, as quantum states of Hermitian operators, and throughout functional analysis. The idea never changes: find the special directions or states that a transformation leaves invariant up to scaling.

Concluding thoughts

This journey traced a single thread. A matrix *transforms* space; the **determinant** measures how much it scales volume and whether it can be undone, while the **trace** sums the diagonal. Together they are the sum and product of the **eigenvalues**, which the **characteristic equation** $\det(A - \lambda I) = 0$ hunts down — and the **eigenvectors** are the directions a transformation cannot turn. For **symmetric** matrices the **Spectral Theorem** hands us a perfect set of perpendicular eigen-axes and the decomposition $A = Q\Lambda Q^\top$, generalised through **Hermitian** matrices and made constructive by **Cholesky**. These are not arrays of numbers; they are operators with a hidden skeleton — and seeing that skeleton is what powers PCA, SVD, simulation, dynamics, and quantum theory.